

PLAN: Discrete Time Markov Chains

- Markov Property, Transition Probability Matrix.
- Time Homogeneous DTMC - Chapman Kolmogorov Equation.

REFERENCE: Chapter 2, "Discrete Event Stochastic Processes", A. Kumar, IISc.

Conditional Independence

Given a prob. space $(\Omega, \mathcal{F}, P)$, two events A and $B \in \mathcal{F}$ are said to be conditionally independent given another event $C \in \mathcal{F}$ s.t. $P(C) > 0$, denoted $A \perp\!\!\!\perp B | C$ if $P(A \cap B | C) = P(A | C) P(B | C)$ or equivalently

$$P(A | B, C) = P(A | C)$$

For $C \in \mathcal{F}$ s.t. $P(C) > 0$ define $P_C: \mathcal{F} \rightarrow [0, 1]$ as $P_C(A) = P(A | C)$ for $A \in \mathcal{F}$

Is this a prob. measure on $(\Omega, \mathcal{F}, P)$? YES!

Example: The RV $X_0 \sim \text{unif} \{0.1, 0.9\}$. When X_0 takes value $p \in \{0.1, 0.9\}$

X_1 and X_2 are IID $\text{Ber}(p)$ RVs.

Define the events $A = \{X_1 = 1\}$, $B = \{X_2 = 1\}$, $C = \{X_0 = 0.1\}$

Is $A \perp\!\!\!\perp B$? Is $A \perp\!\!\!\perp B | C$?

$$P(A \cap B) = P(A \cap B | C) \cdot P(C) + P(A \cap B | C^c) \cdot P(C^c)$$

$$= (0.1)^2 \cdot \frac{1}{2} + (0.9)^2 \cdot \frac{1}{2} = 0.41$$

$$P(A)P(B) = \left[P(A | C) \cdot P(C) + P(A | C^c) \cdot P(C^c) \right] \left[P(B | C) \cdot P(C) + P(B | C^c) \cdot P(C^c) \right]$$
$$(0.1 \cdot \frac{1}{2} + 0.9 \cdot \frac{1}{2}) (0.1 \cdot \frac{1}{2} + 0.9 \cdot \frac{1}{2}) = 0.25$$

5x5 Snakes and Ladders

1	2	3	4	5
10	9	8	7	6
11	12	13	14	15
20	19	18	17	16
21	22	23	24	25

Four die $\{X_n\}_{n \geq 1}$

X_n - state which you are currently on.

$S = \{1, \dots, 25\}$ - State Space

$X_3 = 4$

$X_1 = 1, X_2 = 2, X_3 = 4$

$X_4 \in \{5, 17, 7, 8, 9, 10\}$

$$P(X_4 = 7 | X_1 = 1, X_2 = 2, X_3 = 4) \\ = P(X_4 = 7 | X_3 = 4)$$

Given the present, the future is independent of the past.

The Markov Property

Fix a prob. space $(\Omega, \mathcal{F}, \mathbb{P})$

Defn: A process $\{X_t : t \in \mathcal{T}\}$ is said to be a Markov process if for any

$$t \in \mathcal{T} \quad \underbrace{(X_s : s < t)}_{\text{Past } < t} \perp \underbrace{(X_s : s > t)}_{\text{Future } > t} \mid X_t$$

$\xleftarrow{s < t} \quad t \quad \xrightarrow{s > t}$
 present

$\forall m, n \in \mathbb{N}, s_1, s_2, \dots, s_m, u_1, \dots, u_n \in \mathcal{T}$

$$P(X_{s_1} \leq x_1, X_{s_2} \leq x_2, \dots, X_{s_m} \leq x_m, X_{u_1} \leq y_1, X_{u_2} \leq y_2, \dots, X_{u_n} \leq y_n \mid X_t = x) \\ = P(X_{s_1} \leq x_1, X_{s_2} \leq x_2, \dots, X_{s_m} \leq x_m \mid X_t = x) \cdot P(X_{u_1} \leq y_1, \dots, X_{u_n} \leq y_n \mid X_t = x)$$

Part (before t) Future (after t)

Discrete time Markov Chain

$$X_1 \rightarrow X_2 \rightarrow X_3 \rightarrow X_4 \dots$$

Fix a prob. space $(\Omega, \mathcal{F}, \mathbb{P})$

Defn: A random process $\{X_n, n \in \mathbb{N}\}$ taking values in a countable set S is called a discrete time Markov Chain (DTMC) if it has the Markov property, i.e., for all $i_1, \dots, i_{n-1}, i_n, j \in S$ and $n \in \mathbb{N}$

$$\mathbb{P}\{X_{n+1}=j \mid X_1=i_1, \dots, X_{n-1}=i_{n-1}, X_n=i_n\} = \mathbb{P}\{X_{n+1}=j \mid X_n=i_n\}$$

The set S is called the state space.

Example: $\{X_n\}_{n \geq 1}$ $X_i \sim \text{iid } \text{Be}(p)$

$$Y_{n+1} \perp\!\!\!\perp \{Y_1, \dots, Y_{n-1}\} \mid Y_n$$
$$Y_{n+1}, Y_{n-1}$$

Markov Property holds trivially

Define $Y_n = \sum_{i=1}^n X_i$; $n \geq 1$

$$Y_{n+1} = Y_n + X_{n+1}$$

$$\mathbb{P}\{Y_{n+1}=j \mid Y_1=i_1, \dots, Y_n=i_n\} = \begin{cases} 0 & \text{if } j \notin \{i_n, i_n+1\} \\ p & \text{if } j = i_n+1 \\ 1-p & \text{if } j = i_n \end{cases}$$
$$= \mathbb{P}(Y_{n+1}=j \mid Y_n=i_n)$$

Thus $\{Y_n\}_{n \geq 1}$ is a DTMC on the state space $\mathbb{N} \cup \{0\}$

Is $Y_{n+1} \perp\!\!\!\perp (Y_1, \dots, Y_{n-1})$? No!
(Y_n is not given)

No guarantee on unconditional independence

Homework: If $\{X_n: n \in \mathbb{N}\}$ is a DTMC on S , then for any $A_1, \dots, A_{n-1} \subset S$ and $i, j \in S$ prove that

$$\mathbb{P}(X_{n+1}=j \mid X_1 \in A_1, X_2 \in A_2, \dots, X_{n-1} \in A_{n-1}, X_n=i) = \mathbb{P}(X_{n+1}=j \mid X_n=i)$$

2) If $\{X_n, n \in \mathbb{N}\}$ is a DTMC on S , then for $n < n_1 < n_2 < \dots < n_m$, and $i_1, i_2, \dots, i_{n-1}, i, j_1, \dots, j_m \in S$, show that

$$\mathbb{P}(X_{n_1} = j_1, X_{n_2} = j_2, \dots, X_{n_m} = j_m \mid X_1 = i_1, \dots, X_{n-1} = i_{n-1}, X_n = i)$$

$$= \mathbb{P}(X_{n_1} = j_1, X_{n_2} = j_2, \dots, X_{n_m} = j_m \mid X_n = i)$$

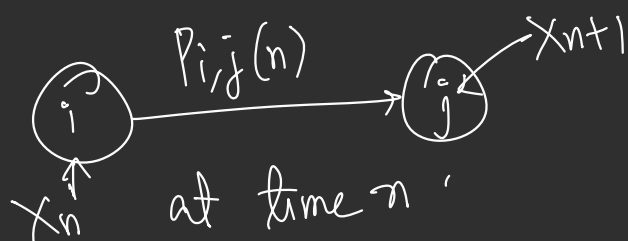
Transition Probability Matrix (TPM)

Let $\{X_n, n \geq 0\}$ be a DTMC on the state space S .
 The first order pmf of X_n is denoted as $\pi_n = (\pi_n(i), i \in S)$
 $S = \{1, \dots, 25\}$
 $\pi_{10} = (\pi_{10}(1), \dots, \pi_{10}(25))$
 $\pi_n(i) = \mathbb{P}(X_n = i)$

$$\mathbb{P}(X_{n+1} = j \mid X_n = i) \neq \mathbb{P}(X_2 = j \mid X_1 = i)$$

The conditional prob. of the form $\mathbb{P}\{X_{n+1} = j \mid X_n = i\}$ are denoted as $P_{i,j}(n)$ and are called **transition probabilities** at time step n .

Defn: Let $\{X_n, n \geq 0\}$ be a DTMC on the state space S . The transition prob. matrix (TPM) of the Markov chain at any time $n \in \mathbb{N} \cup \{0\}$ is an $|S| \times |S|$ matrix $P(n) = [P_{ij}(n)]_{i,j \in S}$ defined on $P_{ij}(n) = \mathbb{P}\{X_{n+1} = j \mid X_n = i\}$ $i, j \in S$.



TPM

$$\begin{bmatrix} & & & & \\ & & & & \\ \hline 0 & 0 & \dots & \dots & 0 & \text{Row } i \\ \hline & & & & & \\ & & & & & \\ & & & & & \end{bmatrix}_{|S| \times |S|}$$

For each $i \in S$, $\sum_{j \in S} P_{ij}(n) = 1$

Row-stochastic Matrix
(Row sum is 1, all entries are non-negative)

$P(n)$ is a row-stochastic matrix $\forall n$

$$\sum_{j \in S} P_{ij}(n) = \mathbb{E}[X_{n+1} | X_n = i]$$

$$\text{TPM} : P \rightarrow \begin{bmatrix} P \end{bmatrix}_{|S| \times |S|} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}_{|S|} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}_{|S|}$$

Row i of TPM $P(n)$ = conditional PMF of X_{n+1} given $X_n = i$.

Snake & ladder

$$\underbrace{P(X_5 = b | X_4 = a)} \rightarrow \underbrace{P(X_{10} = b | X_9 = a)}$$

Time Homogeneous DTMC.

Defn: A DTMC $\{X_n, n \in \mathbb{N}\}$ with state space S & TPM, $\{P(n)\}_{n=1}^{\infty}$ is said to be time-homogeneous if $P(n) = P(n+1) \forall n \in \mathbb{N}$

In this case we denote the common TPM as P .

$$P(X_{n+1} = j | X_n = i) = P(X_2 = j | X_1 = i)$$

$$P(X_1 = i_1, \dots, X_n = i_n) = P(X_1 = i_1) P(X_2 = i_2 | X_1 = i_1) P(X_3 = i_3 | X_2 = i_2) \dots P(X_n = i_n | X_{n-1} = i_{n-1})$$

$$= \pi_1(i_1) p_{i_1, i_2}(1) p_{i_2, i_3}(2) \dots p_{i_{n-1}, i_n}(n-1)$$

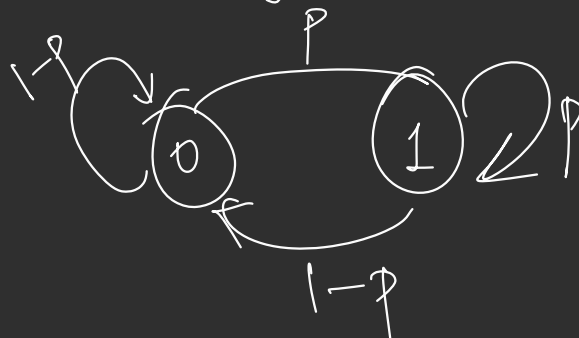
$$= \pi_1(i_1) \prod_{j=1}^{n-1} p_{i_j, i_{j+1}}(j)$$

Initial pmf & the TPM_s together completely specify the process.
 $\{P(n)\}_{n \geq 1}$

For a TH-DTMC, the process is completely specified by π_1 & P .

Example: $\{X_n\}_{n=1}^{\infty} X_n \stackrel{\text{iid}}{\sim} \text{Ber}(p)$ $S = \{0, 1\}$

TPM $P = \begin{bmatrix} 0 & 1 \\ 1-p & p \\ 1 & 1-p \end{bmatrix}$



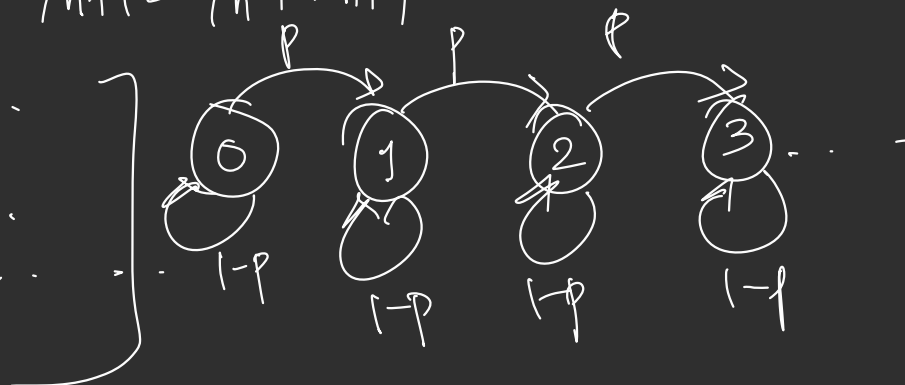
$Y_n = \sum_{i=1}^n X_i$ $\{Y_n\}_{n \geq 1}$

Transition Prob. diagram.

$S = \mathbb{N} \cup \{0\}$

$Y_{n+1} = Y_n + X_{n+1}$

$P = \begin{bmatrix} 0 & 1 & 2 & \dots \\ 0 & 1-p & p & 0 & \dots \\ 1 & 0 & 1-p & p & 0 & \dots \\ 2 & 0 & 0 & 1-p & p & 0 & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}$



n-step transition probabilities:

$P_{ij}^{(n)} \rightarrow$ one-step TP at time instant n
 $= P(X_{n+1}=j | X_n=i)$

$$P\{X_{n+1}=j | X_1=i\} = P_{ij}^{(n)} \leftarrow \begin{array}{l} \text{n-step TP} \\ \text{from } i \text{ to } j \end{array}$$

This is talked about only for TH-DTMCs.

Consider $\{X_n\}_{n \in \mathbb{N}}$ on state space S with TPM P .

$$P\{X_{n+1}=j | X_n=i\} = P_{ij} \quad \forall n \in \mathbb{N}$$

$$P\{X_2=j | X_1=i\} = P_{ij}$$

$$P\{X_3=j | X_1=i\} = \sum_{k \in S} P(X_3=j | X_2=k, X_1=i) \cdot P(X_2=k | X_1=i)$$

$$= \sum_{k \in S} \underbrace{P(X_3=j | X_2=k)}_{\text{1st step}} \cdot \underbrace{P(X_2=k | X_1=i)}_{\text{2nd step}}$$

$$= \sum_{k \in S} P_{kj} \cdot P_{ik}$$

$$= P_{ij}^2$$

$$\begin{bmatrix} \text{ifn} \end{bmatrix} \begin{bmatrix} \text{jth coln} \end{bmatrix}$$

We can get

$$P\{X_4=j | X_1=i\} = \sum_{k \in S} P(X_4=j | X_3=k, X_1=i) \cdot P(X_3=k | X_1=i)$$

Thm: Chapman-Kolmogorov: For a TH-DTMC $\{X_n, n \in \mathbb{N}\}$ with state space S , for any n let $P_{ij}^{(n)} = P\{X_{n+1}=j | X_1=i\}$, $i, j \in S$. Let $P^{(n)} = [P_{ij}^{(n)}]_{i,j \in S}$. Then $P^{(n)} = P^n$, $\forall n \in \mathbb{N}$ where $P = P^{(1)}$ is the one-step TPM of $\{X_n, n \in \mathbb{N}\}$.